Package 'SIMBAH'

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Title SIMBA-H: Supervised Integrative Modeling for Biclustering and Analysis via HMM

Version 0.1.1

Description Implements a Bayesian supervised biclustering framework
using Hidden Markov Models for multi-modality data integration.
The model performs joint clustering and outcome modeling via
Gibbs sampling.

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R topics documented:

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`Cindex` *Concordance Index (C-index) for Censored Outcomes*

Description

Computes the concordance index using predicted times, observed times, and event indicators.

Usage

```
Cindex(Tpredict, Ttest, Status)
```

Arguments

Tpredict	Numeric vector of predicted times.
Ttest	Numeric vector of observed times.
Status	Integer/numeric vector of event indicators (1 = event observed, 0 = censored).

Value

Numeric scalar concordance index.

EvaluateTest *Evaluate Test-Set Predictions*

Description

Computes an outcome-specific performance metric using posterior predictive samples returned by iBiclust.

Usage

```
EvaluateTest(result, TestY, OutcType, threshold = 0, return_all = FALSE)
```

Arguments

result	A fitted model object returned by iBiclust containing YpredTestSample (posterior predictive samples for test outcomes).
TestY	Test-set response variable. For continuous or binary outcomes, this should be a numeric vector. For censored outcomes, this must be an $n \times 2$ matrix of the form cbind(time, censor), where the second column is a censoring indicator (1 = censored, 0 = event observed).
OutcType	Outcome type: one of "continuous", "binary", or "censored".
threshold	Numeric; classification threshold for binary outcomes applied to the posterior predictive mean. Default is 0.
return_all	Logical; if TRUE, returns a list including intermediate predictions. If FALSE, returns only the metric(s).

Value

For OutcType = "continuous": a scalar mean square error MSE. For OutcType = "binary": a scalar misclassification error MCE. For OutcType = "censored": a scalar concordance index Cindex.

Examples

```
## Not run:
## Assume 'result' is a fitted iBiclust model with test predictions
## and 'datTest$Y' is the corresponding test response.

metric <- iEvaluateTest(
  result = result,
  TestY = datTest$Y,
  OutcType = "continuous"
```

```
)

metric

## End(Not run)
```

`iBiclust` *Bayesian Supervised Biclustering with SIMBA-H or SIMBA for Multi-Modal Data*

Description

Fits a Bayesian model-based supervised biclustering model for multi-modality datasets using a Gibbs sampling algorithm. SIMBA-H incorporates a Hidden Markov Model (HMM) structure across ordered features; SIMBA does not incorporate the HMM structure. Each modality/platform is provided as a matrix with samples in rows and features in columns; internally, each matrix is optionally standardized (centered/scaled) and transposed before being passed to the native routine.

Usage

```
iBiclust(
  Method = "SIMBA-H",
  dataList = dataList,
  OutcType = "continuous",
  Youtcome = Youtcome,
  Cov = NULL,
  sample = 2000,
  burnin = 1000,
  K = 2,
  normalized = TRUE,
  seed = 1,
  hyperSigma = c(1, 1),
  alphaP = rep(1, K),
  delt = rep(1, 3),
  TruncVal = 0.1,
  dataListTest = NULL,
  CovTest = NULL,
  return_samples = FALSE
)
```

Arguments

`Method`	Character string specifying the modeling approach: either `"SIMBA-H"` (default), which incorporates a Hidden Markov Model (HMM) structure across ordered features, or `"SIMBA"`, which does not incorporate the HMM structure.
`dataList`	A list of modality matrices. Each `dataList[[t]]` is an $n \times p_t$ matrix with samples in rows and features in columns.
`OutcType`	Outcome type: one of `"continuous"`, `"binary"`, or `"censored"`.
`Youtcome`	Outcome vector/matrix. For censored outcomes, this must be an $n \times 2$ matrix `cbind(time, censor)`, where the second column represents the *censoring indicator* (1 = censored, 0 = event observed).

Cov	Covariate matrix of dimension $n \times p_c$.
sample	Number of MCMC samples to retain.
burnin	Number of burn-in iterations.
K	Number of sample clusters used in the model.
normalized	Logical; if TRUE, each modality is centered and scaled feature-wise.
seed	Integer random seed passed to the native routine.
hyperSigma	Numeric vector of length 2. The prior distribution on the state-specific variances σ^2_{dk1} and σ^2_{dk2} for $\mu_{d,j,k,l}$.
alphaP	Numeric vector of length K specifying the Dirichlet prior concentration parameters for the cluster probability vector. These parameters control the prior probability that a sample belongs to each cluster k.
delt	Numeric vector of length 3 giving the Dirichlet prior concentration parameters $(\delta_1, \delta_2, \delta_3)$. For Method = "SIMBA-H", these specify the prior for each row of the transition matrix ξ. For Method = "SIMBA", they specify the prior for the feature-state probability vector.
TruncVal	Positive numeric truncation threshold t used in the Gibbs sampler.
dataListTest	Optional test set list with same structure as dataList.
CovTest	Optional test covariate matrix of dimension $n_{test} \times p_c$.
return_samples	Logical; additionally return ZSample, probZSample, MuSample, and probKappaSample when TRUE (default FALSE).

Details

Data format:

- dataList must be a list of length M (number of modalities/platforms).
- Each element dataList[[t]] is an $n \times p_t$ matrix: n samples (rows) and p_t features (columns).
- If normalized = TRUE, each modality in dataList is centered and scaled feature-wise using its own mean and standard deviation. If dataListTest is provided, the test data are standardized using the corresponding training means and standard deviations (not re-estimated from the test data).

Ordering requirement (HMM): If Method = "SIMBA-H" (the default), the features within each modality in dataList are assumed to be pre-ordered according to their natural or genomic sequence. The function does not reorder features internally. For Method = "SIMBA", no feature ordering is required.

Outcome types:

- "continuous": OutcomeType = 1
- "binary": OutcomeType = 2
- "censored": OutcomeType = 3, where Youtcome must be an $n \times 2$ matrix (e.g., value and censoring indicator).

Test data: If dataListTest is provided, it must have the same list structure as dataList. If normalized = TRUE, test data are standardized using the training means/SDs.

Value

A list with posterior summaries, coefficient draws, test labels and predictions, and log-posterior values. Only ZSample, probZSample, MuSample, and probKappaSample require `return_samples = TRUE`. Label switching is checked automatically. If label switching occurs, cluster labels are aligned using iterative ECR:

ZSample, probZSample, MuSample, probKappaSample Retained labels, cluster probabilities, latent means, and feature-state probabilities when requested.

DIC Deviance Information Criterion returned by the sampler.

probZMean Matrix of posterior mean cluster membership probabilities ($n \times K$).

MuMean Array of posterior means of the latent mean parameters μ_{dkl} for the over- and under-expressed states ($l = 1, 2$), indexed by modality d and cluster k.

CoefySample cluster-specific outcome means with dimension $sample \times K$.

sig2yM Posterior mean of outcome variance parameter.

KappaMean List of arrays of posterior means for biclustering indicators per modality.

TransMatMean Array of posterior mean transition matrices per modality for `Method = "SIMBA-H"`. For `Method = "SIMBA"`, each matrix has identical rows containing posterior mean feature-state probabilities.

Betasample Matrix of sampled covariate coefficients ($sample \times p_c$).

ZtestSample Matrix of sampled test cluster labels ($sample \times n_{test}$) if test data provided.

YpredTestSample Matrix of sampled test predictions ($sample \times n_{test}$) if test data provided.

logpostsample Vector of log-posterior values of length `sample + burnin`.

Examples

```
## Not run:
## Generate synthetic multi-modality data
K <- 2
p <- c(300, 500)   # smaller for an example
n <- 100
nbrcov <- 4
OutcType <- "continuous"

dat <- iGenDataHMM(
  OrderFeatured = TRUE, nActive = 100, K = K, NbrPlat = 2, OutcType = OutcType,
  MeanY = c(rep(1, floor(K/2)), rep(-1, K - floor(K/2))),
  pc = nbrcov, betacov = 1.0,
  MeanOverExpre = (1:K) / 2, MeanUnderExpre = -(1:K) / 2,
  p = p, seed = 1, n = n,
  SigmaOE = rep(1, K), SigmaUE = rep(1, K)
)

## Generate test data
ntest <- 100
datTest <- iGenDataHMM(
  OrderFeatured = TRUE, nActive = 100, K = K, NbrPlat = 2, OutcType = OutcType,
  MeanY = c(rep(1, floor(K/2)), rep(-1, K - floor(K/2))),
  pc = nbrcov, betacov = 1.0,
  MeanOverExpre = (1:K) / 2, MeanUnderExpre = -(1:K) / 2,
  p = p, seed = 2, n = ntest,
  SigmaOE = rep(1, K), SigmaUE = rep(1, K)
```

```
)

## Fit SIMBA-H with the HMM structure (keep MCMC small in examples)
## Use Method = "SIMBA" to fit without the HMM structure.
result <- iBiclust(
  Method = "SIMBA-H",
  dataList = dat$X,
  Youtcome = dat$Y,
  OutcType = OutcType,
  Cov = dat$Cov,
  sample = 200,
  burnin = 100,
  K = K,
  normalized = FALSE,
  seed = 1,
  hyperSigma = c(1, 1),
  alphaP = rep(1, K),
  delt = rep(1, 3),
  TruncVal = 0.1,
  dataListTest = datTest$X,
  CovTest = datTest$Cov
)

str(result)

## End(Not run)
```

---

iGenDataHMM *Generate Synthetic Multi-Modal Data for HMM-Based Supervised Biclustering*

Description

Simulates multi-modality feature matrices, covariates, and an outcome under a cluster/state structure used by iBiclust. For each modality, features are assigned to one of three states within each cluster: over-expressed (1), under-expressed (2), or baseline (3). Outcomes can be continuous, binary, or censored.

Usage

```
iGenDataHMM(
  OrderFeatured = FALSE,
  K = K,
  NbrPlat = 2,
  OutcType = "continuous",
  MeanY = MeanY,
  pc = NULL,
  betacov = 1,
  MeanOverExpre = 1:K,
  MeanUnderExpre = -c(1:K),
  p = p,
  seed = 1,
```

```
  n = n,
  SigmaOE = rep(1, K),
  SigmaUE = rep(1, K),
  nActive = 100L
)
```

Arguments

OrderFeatured	Logical; if TRUE, uses a deterministic ordering for feature membership in each modality (the first nActive features over-expressed and the next nActive under-expressed). If FALSE, feature membership is randomized.
K	Integer; number of clusters.
NbrPlat	Integer; number of modalities/platforms/views.
OutcType	Outcome type: one of "continuous", "binary", or "censored".
MeanY	Numeric scalar, length-K vector, or NbrPlat by K matrix; cluster-specific mean outcome levels.
pc	Integer; number of covariates to generate. Must be a positive integer.
betacov	Numeric; magnitude of the covariate coefficients. Coefficients are assigned as +(betacov) for the first half and -(betacov) for the remainder.
MeanOverExpre	Numeric scalar, length-K vector, or NbrPlat by K matrix; means for over-expressed features in each cluster.
MeanUnderExpre	Numeric scalar, length-K vector, or NbrPlat by K matrix; means for under-expressed features in each cluster.
p	Integer vector of length NbrPlat; number of features per modality.
seed	Integer; random seed used for simulating X and outcomes.
n	Integer; number of samples.
SigmaOE	Numeric scalar, length-K vector, or NbrPlat by K matrix; standard deviations for over-expressed features.
SigmaUE	Numeric scalar, length-K vector, or NbrPlat by K matrix; standard deviations for under-expressed features.
nActive	Positive integer or length-NbrPlat vector; number of features in each active state per modality (default 100).

Details

Cluster assignment: Samples are assigned to K clusters in approximately equal-sized blocks.

Feature membership (kappajk): For each modality m, a $p_m \times K$ matrix is generated with entries in {1,2,3}. If OrderFeatured = TRUE, the first nActive features are set to state 1 (over-expressed) and the next nActive features to state 2 (under-expressed). Otherwise, nActive over- and nActive under-expressed features are sampled at random for each modality (requires p[m] >= 2 * nActive[m]).

Outcome generation: A latent mean MeanY[Zk[i]] is assigned to each sample and combined with a linear covariate effect. Continuous outcomes add Gaussian noise; binary outcomes threshold a noisy latent score; censored outcomes return an $n \times 2$ matrix cbind(time, censor), where the second column is a censoring indicator (1 = censored, 0 = event observed).

Value

A list with components:

Y Outcome vector of length `n` for continuous/binary outcomes, or an $n \times 2$ matrix for censored outcomes `cbind(time, censor)`, where the second column is the censoring indicator (1 = censored, 0 = event observed).

X List of length `NbrPlat`; each element is an $n \times p_m$ matrix of simulated features.

kappajk List of length `NbrPlat`; each element is a $p_m \times K$ matrix with entries in `{1,2,3}` indicating over/under/baseline feature membership for each cluster.

Zk Integer vector of length `n` giving true cluster assignments in `1:K`.

Cov Numeric matrix of dimension $n \times pc$ containing generated covariates.

Examples

```
set.seed(1)
K <- 2
p <- c(300, 500)
n <- 100
dat <- iGenDataHMM(
  OrderFeatured = TRUE, nActive = 100, K = K, NbrPlat = 2,
  OutcType = "continuous", MeanY = c(1, -1), pc = 4, betacov = 1.0,
  MeanOverExpre = (1:K) / 2, MeanUnderExpre = -(1:K) / 2,
  p = p, seed = 1, n = n,
  SigmaOE = rep(1, K), SigmaUE = rep(1, K)
)
str(dat)
```

ShowBicluterMeanTrace *Plot MCMC Traces of Bicluster Means*

Description

Creates trace plots for the overexpressed and underexpressed bicluster means retained by `iBiclust`. The plot shows retained iteration on the horizontal axis and mean parameter value on the vertical axis, with colors identifying clusters and panels identifying modalities and expression states.

Usage

```
ShowBicluterMeanTrace(result)
```

Arguments

`result`	Results returned by `iBiclust` with `return_samples = TRUE`. Its `MuSample` component must be an array with dimensions retained iterations by modalities by clusters by expression states, where state 1 is overexpressed and state 2 is underexpressed.

Value

A list with two components:

data A long-format data frame containing the retained iteration `id`, cluster `k`, modality label `d`, expression state `name`, sampled mean `value`, modality/cluster label `group`, and `chain` (set to 1).

figure A `ggplot` object. Use `print()` to display it.

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